

DIGISIM PS-1

Team Name- Kaizer

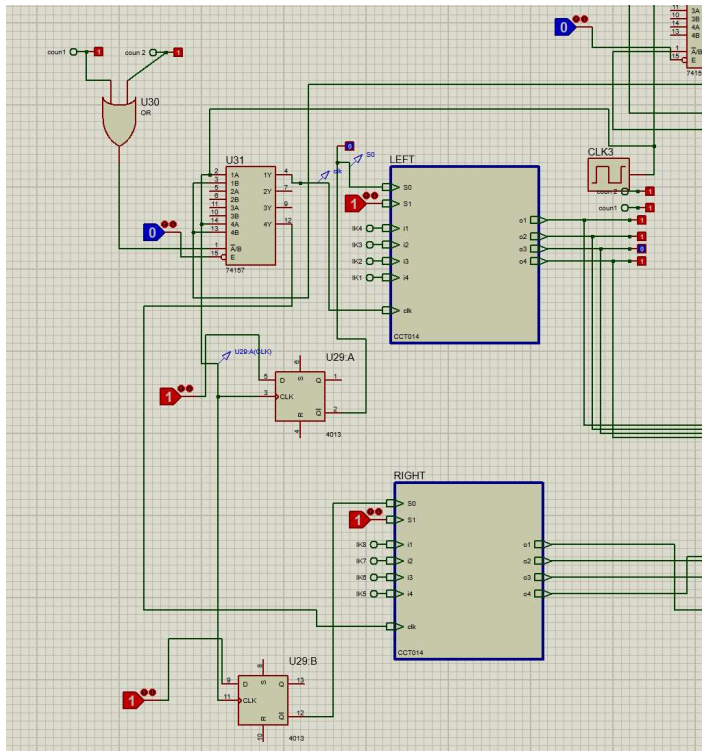
Member 1- Pratyush Meshram

Member 2- Aditya Kishore

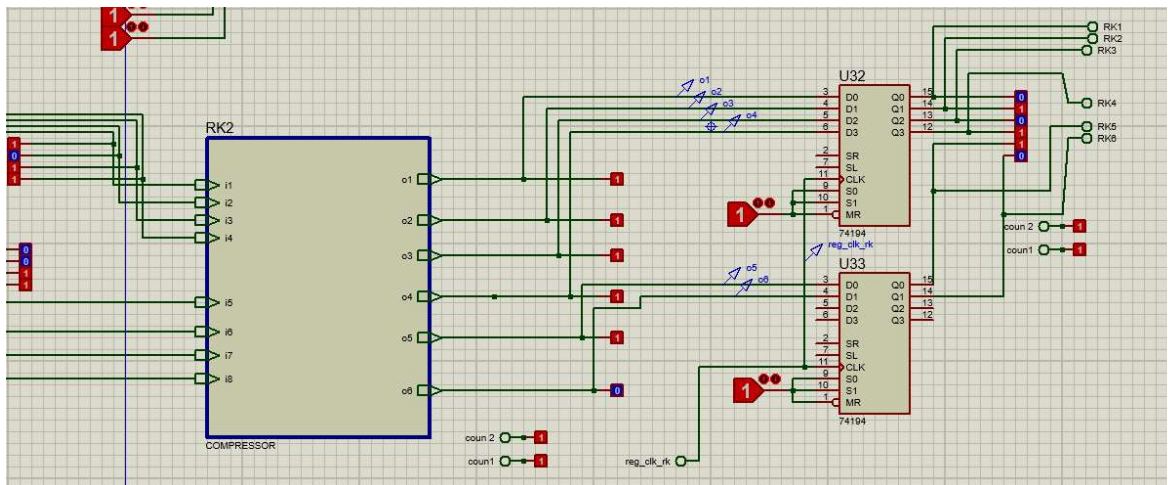
Sequential Circuit approach(Cost= Rs. 99.3)-

Round key generation:

- To generate a round key, we divide it into left and right halves first. Then, we give it a single pulse for each even counter output value (apart from the initial one, since loading is also required here, so we first need a double pulse) and a double pulse (double the frequency of the initial counter) for each odd counter output. Finally, we add them using a 2:1 MUX to create the waveform, or clock signal:

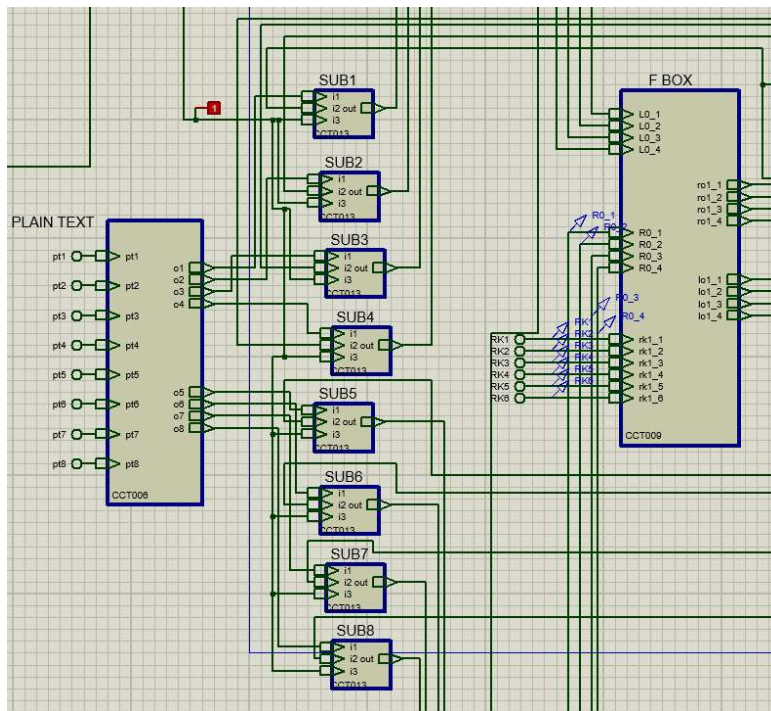


- Next, connect each one to a left circular shift register separately. Subsequently, supply the left and right results to the Compressor block (made like the way in combinational method). Lastly, supply the 6-bit round key, which we store in a register at a clock frequency equal to the input counter frequency.

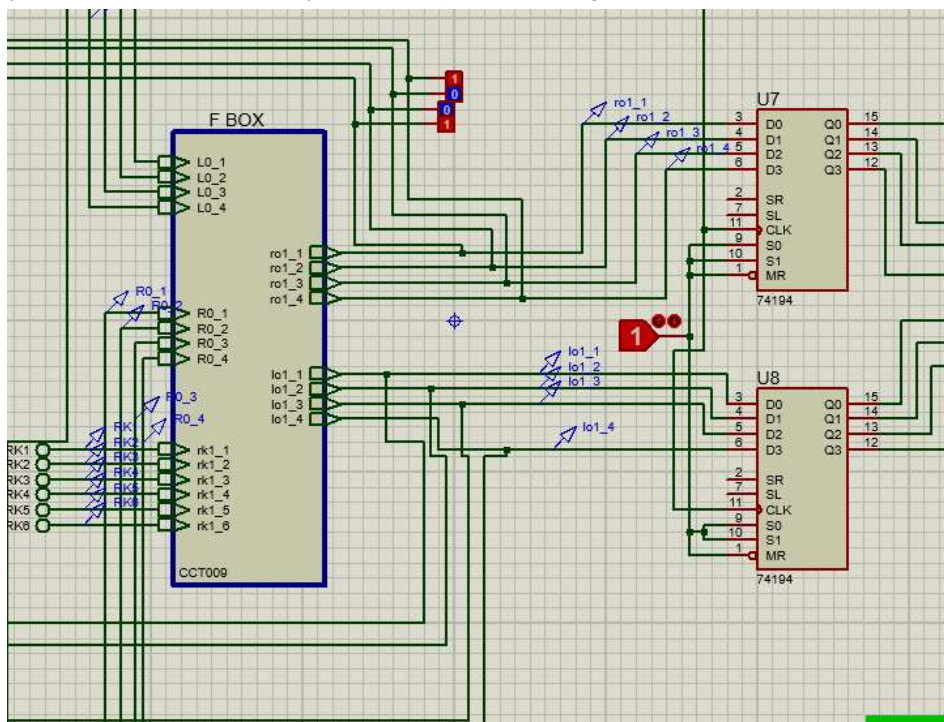


Function generator:

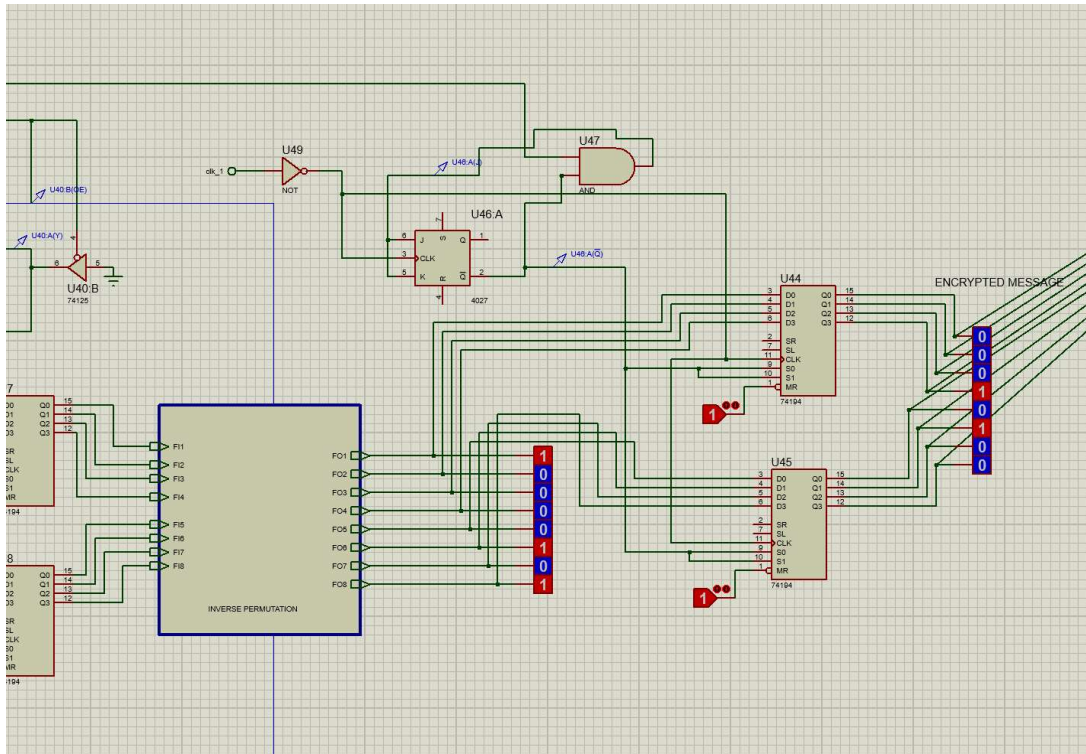
- First, we run our simple 8-bit text through a permutator block. Next, we send each bit via a 2:1 MUX, passing either the feedback from the previous round or the plain text for the first iteration, and then we pass to the F-BOX (same as in combinational), which completes the round fn.



- Next, we supply them to the F- BOX, which is the function (f) and XOR gate combined. It generates R+1 and L+1 states by taking an 8-bit text and a 6-bit key as input.
- After that it takes the output as the 4 bits which have been taken out from the S-boxes (same as combinational) and then permuted again.

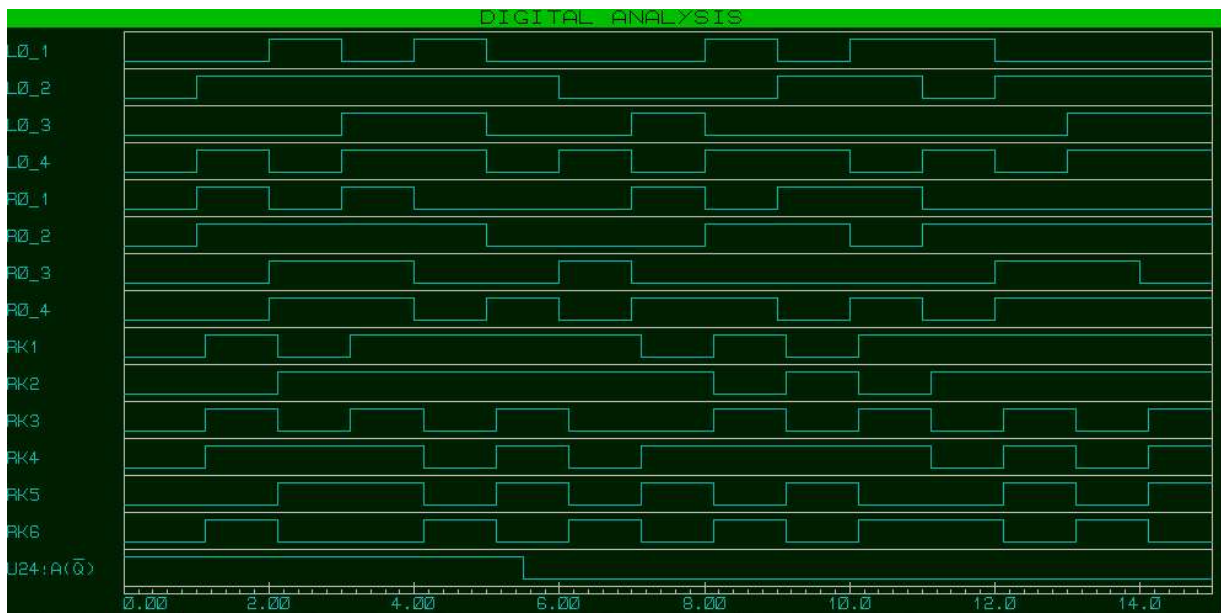


- Then, we pass it to the encrypted block for inverse permutation. At this point, we want to stop the changing output after 4 iterations, which is easily reset to a different number. To do this, we store the values in a register, the load of which will change using a T-flip flop once the counter reaches the fifth iteration.



The final output, or encrypted code, is obtained when the counter reaches the fifth iteration.

CLOCK PULSES:



Cost:

- 1 clock
Cost= Rs.40
- Round key generator:
 - 3 gates
 - 3 MUX
 - 4 Registers
 - 2 Counters
 - 4 Flip flopsCost= Rs. 24.3
- Function generator:
 - 2 3-input gates, 26 2-input gates
 - 14 registers
 - 1 counter
 - 2 Flip flopsCost= Rs. 35

Total Cost= Rs. 99.3